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ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Hanoi, July 20, 2026

To: Graduate Admissions Committee
Department of Electrical and Computer Engineering
University of Michigan–Dearborn
4901 Evergreen Road, Dearborn, MI 48128, USA
Subject: Letter of Recommendation for Xuan Thanh Trinh

Dear Members of the Admissions Committee,

I am Nguyen Duc Tuyen, Head of the Power Grid and Renewable Energy (PGRE) Laboratory at Hanoi University of Science and Technology (HUST). I also serve as an Adjunct Associate Professor at the Shibaura Institute of Technology (Japan), where I earned my PhD in Electrical Engineering. Over the past decade, my research has focused on smart grids, microgrids, and the integration of renewable energy systems, leading to over 100 scientific publications.

I am writing to strongly recommend my student, Xuan Thanh Trinh, for admission to the PhD program at the University of Michigan-Dearborn. Among the students I have advised, Thanh stands out as a exceptionally autonomous and analytical undergraduate research assistant. He has joined my research group for two years with a clear interest in using mathematical optimization to address renewable energy integration. From the beginning, he demonstrated a methodical approach to his work. Specifically, he proactively synthesized literature to identify critical research gaps and translated complex mathematical models into functional code that seamlessly combines established computational tools to complement novel frameworks. Notably, when encountering complex technical errors where other lab members were unable to provide technical guidance, he independently analyzed the issues and refined his approach. Moreover, he consistently sought to improve his initial results, ensuring his solutions met practical grid requirements. This persistence and dedication to problem solving led directly to a successful oral presentation at the ICGEA conference and an excellent Bachelor's thesis.

Thanh's strong discipline and time management are evident in his ability to balance research, coursework, and industry practice. While actively contributing to our lab, he consistently ranked in the top 3 of his class, demonstrating a solid understanding of Power System principles. He further validated this theoretical knowledge through hands-on experience during his internship at the Northern Electrical Testing Company (ETC1). Successfully managing these responsibilities highlights the capability and strong sense of responsibility he will bring to your PhD program.

Building on these qualities, Thanh has also shown a natural ability to guide and support others within our laboratory. Rather than focusing solely on his own projects, he proactively built automated workflows to assist the entire team. He also readily dedicates his time to supporting new members, patiently guiding them through algorithmic modeling. This collaborative mindset has fostered a positive environment and contributed to the lab's overall productivity.

Considering his academic background, research experience, and practical skills, I believe Thanh is well prepared for graduate studies. I am pleased to recommend him for admission to the PhD program at the University of Michigan-Dearborn. If you require any further information, please feel free to contact me.

Sincerely,

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